

**Williams syndrome**

Congenital microdeletion of long arm of chromosome 7 (deleted region includes elastin gene). Findings: distinctive “elfin” facies, intellectual disability, hypercalcemia, well-developed verbal skills, extreme friendliness with strangers, cardiovascular problems (eg, supraventricular aortic stenosis, pulmonary artery stenosis, renal artery stenosis).

## ► BIOCHEMISTRY—NUTRITION

**Essential fatty acids**

Polyunsaturated fatty acids that cannot be synthesized in the body and must be provided in the diet (eg, nuts/seeds, plant oils, seafood). Linoleic acid (omega-6) is metabolized to arachidonic acid, which serves as the precursor to leukotrienes and prostaglandins. Linolenic acid (omega-3) and its metabolites have cardioprotective and antihyperlipidemic effects.

In contrast, consumption of *trans*-unsaturated fatty acids (found in fast food) promotes cardiovascular disease by ↑ LDL and ↓ HDL.

**Vitamins: fat soluble**

A, D, E, K. Absorption dependent on bile emulsification, pancreatic secretions, and intact ileum. Toxicity more common than for water-soluble vitamins because fat-soluble vitamins accumulate in fat.

Malabsorption syndromes with steatorrhea (eg, cystic fibrosis and celiac disease) or mineral oil intake can cause fat-soluble vitamin deficiencies.

**Vitamins: water soluble**

B<sub>1</sub> (thiamine: TPP)  
B<sub>2</sub> (riboflavin: FAD, FMN)  
B<sub>3</sub> (niacin: NAD<sup>+</sup>)  
B<sub>5</sub> (pantothenic acid: CoA)  
B<sub>6</sub> (pyridoxine: PLP)  
B<sub>7</sub> (biotin)  
B<sub>9</sub> (folate)  
B<sub>12</sub> (cobalamin)  
C (ascorbic acid)

Wash out easily from body except B<sub>12</sub> and B<sub>9</sub>. B<sub>12</sub> stored in liver for ~ 3–4 years. B<sub>9</sub> stored in liver for ~ 3–4 months. B-complex deficiencies often result in dermatitis, glossitis, and diarrhea. Can be coenzymes (eg, ascorbic acid) or precursors to coenzymes (eg, FAD, NAD<sup>+</sup>).

**Dietary supplementation**

| DIET                 | SUPPLEMENTATION REQUIRED                                                                                                               |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Vegetarian/vegan     | Vitamin B <sub>12</sub><br>Iron<br>Vitamin B <sub>2</sub><br>Frequently, vitamin D (although this is commonly deficient in many diets) |
| High egg white (raw) | Vitamin B <sub>7</sub> (avidin in egg whites binds biotin and prevents absorption)                                                     |
| Untreated corn       | Vitamin B <sub>3</sub> (deficiency is common in resource-limited areas)                                                                |

**Vitamin A**

Includes retinal, retinol, retinoic acid.

## FUNCTION

Antioxidant; constituent of visual pigments (**retinal**); essential for normal differentiation of epithelial cells into specialized tissue (pancreatic cells, mucus-secreting cells); prevents squamous metaplasia.

**Retinol** is vitamin **A**, so think **retin-A** (used topically for wrinkles and **Acne**).  
Found in liver and leafy vegetables.  
Supplementation in vitamin A-deficient measles patients may improve outcomes.  
Use oral isotretinoin to treat severe cystic acne.  
Use *all-trans* retinoic acid to treat acute promyelocytic leukemia.

## DEFICIENCY



Night blindness (nyctalopia); dry, scaly skin (xerosis cutis); dry eyes (xerophthalmia); conjunctival squamous metaplasia → Bitot spots (keratin debris; foamy appearance on conjunctiva **A**); corneal degeneration (keratomalacia); immunosuppression.

## EXCESS

Acute toxicity—nausea, vomiting, ↑ ICP (eg, vertigo, blurred vision).  
Chronic toxicity—alopecia, dry skin (eg, scaliness), hepatic toxicity and enlargement, arthralgias, and idiopathic intracranial hypertension.

Teratogenic (interferes with homeobox gene; cleft palate, cardiac abnormalities), therefore a ⊖ pregnancy test and two forms of contraception are required before isotretinoin (vitamin A derivative) is prescribed.  
**Isotretinoin is teratogenic.**

**Vitamin B<sub>1</sub>**

Also called thiamine.

## FUNCTION

In thiamine pyrophosphate (TPP), a cofactor for several dehydrogenase enzyme reactions (**Be APT**):

- **B**ranched-chain ketoacid dehydrogenase
- **α**-Ketoglutarate dehydrogenase (TCA cycle)
- **P**yruvate dehydrogenase (links glycolysis to TCA cycle)
- **T**ransketolase (HMP shunt)

## DEFICIENCY

Impaired glucose breakdown → ATP depletion worsened by glucose infusion; highly aerobic tissues (eg, brain, heart) are affected first. In patients with chronic alcohol overuse or malnutrition, give thiamine before dextrose to ↓ risk of precipitating Wernicke encephalopathy.  
Diagnosis made by ↑ in RBC transketolase activity following vitamin B<sub>1</sub> administration.

## DISORDER

## CHARACTERISTICS

|                                    |                                                                                                                                                                          |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Wernicke encephalopathy</b>     | Acute, reversible, life-threatening neurologic condition. Symptoms: <b>C</b> onfusion, <b>O</b> phthalmoplegia/ <b>N</b> ystagmus, <b>A</b> taxia ( <b>CoRONA</b> beer). |
| <b>Korsakoff syndrome</b>          | Amnesic disorder due to chronic alcohol overuse; presents with confabulation, personality changes, memory loss (permanent).                                              |
| <b>Wernicke-Korsakoff syndrome</b> | Damage to medial dorsal nucleus of thalamus, mammillary bodies. Presentation is combination of Wernicke encephalopathy and Korsakoff syndrome.                           |
| <b>Dry beriberi</b>                | Polyneuropathy, symmetric muscle wasting.                                                                                                                                |
| <b>Wet beriberi</b>                | High-output cardiac failure (due to systemic vasodilation).                                                                                                              |

Spell beriberi as **Ber1Ber1** to remember vitamin **B<sub>1</sub>**.

**Vitamin B<sub>2</sub>**

Also called riboflavin.

## FUNCTION

Component of flavins FAD and FMN, used as cofactors in redox reactions, eg, the succinate dehydrogenase reaction in the TCA cycle.

FAD and FMN are derived from ribo**F**lavin (B<sub>2</sub> ≈ 2 ATP).

## DEFICIENCY

Cheilosis **A** (inflammation of lips, scaling and fissures at the corners of the mouth), “magenta” tongue, corneal vascularization.

The 2 **C**’s of B<sub>2</sub>.

**Vitamin B<sub>3</sub>**

Also called niacin, nicotinic acid.

## FUNCTION

Constituent of NAD<sup>+</sup>, NADP<sup>+</sup> (used in redox reactions and as cofactor by dehydrogenases). Derived from tryptophan. Synthesis requires vitamins B<sub>2</sub> and B<sub>6</sub>. Used to treat dyslipidemia (↓ VLDL, ↑ HDL).

NAD derived from **N**iacin (B<sub>3</sub> ≈ 3 ATP).

## DEFICIENCY

Glossitis. Severe deficiency of B<sub>3</sub> leads to pellagra, which can also be caused by Hartnup disease, malignant carcinoid syndrome (↑ tryptophan metabolism → ↑ serotonin synthesis), and isoniazid (↓ vitamin B<sub>6</sub>). Symptoms of B<sub>3</sub> deficiency (pellagra) (the 3 **D**’s): **d**iarrhea, **d**ementia (also hallucinations), **d**ermatitis (C3/C4 dermatome circumferential “broad collar” rash [Casal necklace], hyperpigmentation of sun-exposed limbs **A**).

**Hartnup disease**—autosomal recessive.

Deficiency of neutral amino acid (eg, tryptophan) transporters in proximal renal tubular cells and on enterocytes → neutral aminoaciduria and ↓ absorption from the gut → ↓ tryptophan for conversion to niacin → pellagra-like symptoms. Treat with high-protein diet and nicotinic acid.

Pellagra = vitamin B<sub>3</sub> levels **fell**.



## EXCESS

Facial flushing (induced by prostaglandin, not histamine; can avoid by taking aspirin before niacin), pruritus, hyperglycemia, hyperuricemia.

Podagra = vitamin B<sub>3</sub> **OD** (overdose).

**Vitamin B<sub>5</sub>**Also called pantothenic acid. B<sub>5</sub> is “**pento**”thenic acid.

## FUNCTION

Component of coenzyme A (CoA, a cofactor for acyl transfers) and fatty acid synthase.

## DEFICIENCY

Dermatitis, enteritis, alopecia, adrenal insufficiency may lead to burning sensation of feet (“burning feet syndrome”; distal paresthesias, dysesthesia).

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Vitamin B<sub>6</sub></b> | Also called pyridoxine.                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                  |
| FUNCTION                     | Converted to pyridoxal phosphate (PLP), a cofactor used in transamination (eg, ALT and AST), decarboxylation reactions, glycogen phosphorylase. Synthesis of glutathione, cystathionine, heme, niacin, histamine, and neurotransmitters including serotonin, epinephrine, norepinephrine (NE), dopamine, and GABA.                                                                                                                     |                                                                                                                                                                                                                                                                                                                  |
| DEFICIENCY                   | Convulsions, hyperirritability, peripheral neuropathy (deficiency inducible by isoniazid and oral contraceptives), sideroblastic anemia (due to impaired hemoglobin synthesis and iron excess).                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                  |
| <b>Vitamin B<sub>7</sub></b> | Also called biotin.                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                  |
| FUNCTION                     | Cofactor for carboxylation enzymes (which add a 1-carbon group): <ul style="list-style-type: none"> <li>Pyruvate carboxylase (gluconeogenesis): pyruvate (3C) → oxaloacetate (4C)</li> <li>Acetyl-CoA carboxylase (fatty acid synthesis): acetyl-CoA (2C) → malonyl-CoA (3C)</li> <li>Propionyl-CoA carboxylase (fatty acid oxidation and branched-chain amino acid breakdown): propionyl-CoA (3C) → methylmalonyl-CoA (4C)</li> </ul> |                                                                                                                                                                                                                                                                                                                  |
| DEFICIENCY                   | Relatively rare. Dermatitis, enteritis, alopecia. Caused by long-term antibiotic use or excessive ingestion of raw egg whites.<br>“ <b>A</b> vidin in egg whites <b>avidly</b> binds biotin.”                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                  |
| <b>Vitamin B<sub>9</sub></b> | Also called folate.                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                  |
| FUNCTION                     | Converted to tetrahydrofolic acid (THF), a coenzyme for 1-carbon transfer/methylation reactions.<br>Important for the synthesis of nitrogenous bases in DNA and RNA.                                                                                                                                                                                                                                                                   | Found in leafy green vegetables. Also produced by gut microbiota. <b>F</b> olate absorbed in <b>jejunum</b> (think <b>f</b> oliage in the “ <b>j</b> ejun”gle).<br>Small reserve pool stored primarily in the liver.                                                                                             |
| DEFICIENCY                   | Macrocytic, megaloblastic anemia; hypersegmented polymorphonuclear cells (PMNs); glossitis; no neurologic symptoms (as opposed to vitamin B <sub>12</sub> deficiency).<br>Labs: ↑ homocysteine, normal methylmalonic acid levels. Seen in chronic alcohol overuse and in pregnancy.                                                                                                                                                    | Deficiency can be caused by several drugs (eg, phenytoin, trimethoprim, methotrexate).<br>Supplemental folic acid at least 1 month prior to conception and during pregnancy to ↓ risk of neural tube defects. Give vitamin B <sub>9</sub> for the <b>9</b> months of pregnancy, and 1 month prior to conception. |

**Vitamin B<sub>12</sub>**

Also called cobalamin.

**FUNCTION**

Cofactor for methionine synthase (transfers CH<sub>3</sub> groups as methylcobalamin) and methylmalonyl-CoA mutase. Important for DNA synthesis.

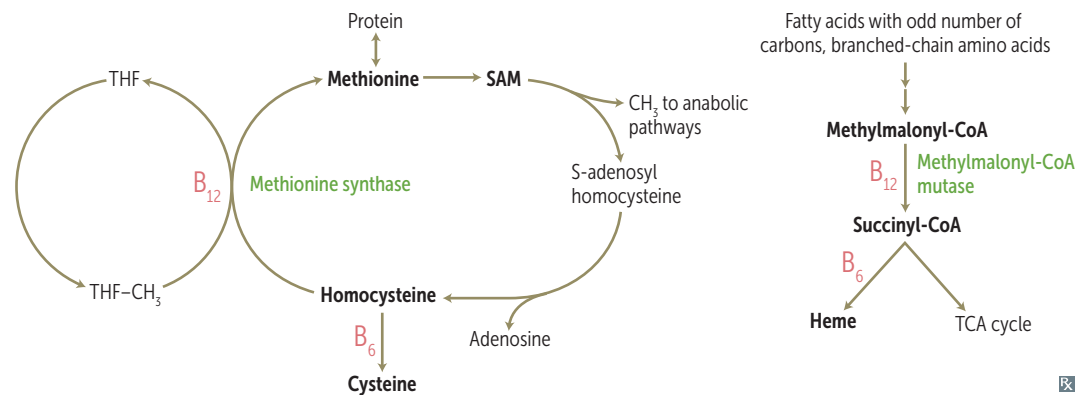
Found in animal products. Synthesized only by intestinal microbiota. Site of synthesis in humans is distal to site of absorption; thus B<sub>12</sub> must be consumed via animal products.

**DEFICIENCY**

Macrocytic, megaloblastic anemia; hypersegmented PMNs; paresthesias and subacute combined degeneration (degeneration of dorsal columns, lateral corticospinal tracts, and spinocerebellar tracts) due to abnormal myelin. Associated with ↑ serum homocysteine and methylmalonic acid levels, along with 2° folate deficiency. Prolonged deficiency → irreversible nerve damage.

Very large reserve pool (several years) stored primarily in the liver. Deficiency caused by malabsorption (eg, sprue, enteritis, *Diphyllobothrium latum*, achlorhydria, bacterial overgrowth, alcohol overuse), lack of intrinsic factor (eg, pernicious anemia, gastric bypass surgery), absence of terminal ileum (surgical resection, eg, for Crohn disease), certain drugs (eg, metformin), or insufficient intake (eg, veganism).

B<sub>9</sub> (folate) supplementation can mask the hematologic symptoms of B<sub>12</sub> deficiency, but not the neurologic symptoms.

**Vitamin C**

Also called ascorbic acid.

**FUNCTION**

Antioxidant; also facilitates iron absorption by reducing it to Fe<sup>2+</sup> state. Necessary for hydroxylation of proline and lysine in collagen synthesis. Necessary for dopamine β-hydroxylase (converts dopamine to NE).

Found in fruits and vegetables.

Pronounce “**absorbic**” acid.

Ancillary treatment for methemoglobinemia by reducing Fe<sup>3+</sup> to Fe<sup>2+</sup>.

**DEFICIENCY**

**Scurvy**—swollen gums, easy bruising, petechiae, hemarthrosis, anemia, poor wound healing, perifollicular and subperiosteal hemorrhages, “corkscrew” hair. Weakened immune response.

Deficiency may be precipitated by tea and toast diet.

Vitamin **C** deficiency causes **sCurvy** due to a **C**ollagen hydro**C**ylation defect.

**EXCESS**

Nausea, vomiting, diarrhea, fatigue, calcium oxalate nephrolithiasis (excess oxalate from vitamin C metabolism). Can ↑ iron toxicity in predisposed individuals by increasing dietary iron absorption (ie, can worsen hemochromatosis or transfusion-related iron overload).

**Vitamin D**

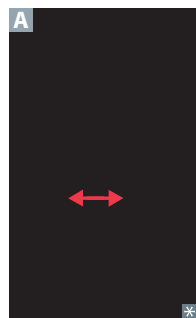
D<sub>3</sub> (cholecalciferol) from exposure of skin (stratum basale) to sun, ingestion of fish, milk, plants.  
 D<sub>2</sub> (ergocalciferol) from ingestion of plants, fungi, yeasts.  
 Both converted to 25-OH D<sub>3</sub> (storage form) in liver and to the active form 1,25-(OH)<sub>2</sub> D<sub>3</sub> (calcitriol) in kidney.

**FUNCTION**

↑ intestinal absorption of Ca<sup>2+</sup> and PO<sub>4</sub><sup>3-</sup>.  
 ↑ bone mineralization at low levels.  
 ↑ bone resorption at higher levels.

**REGULATION**

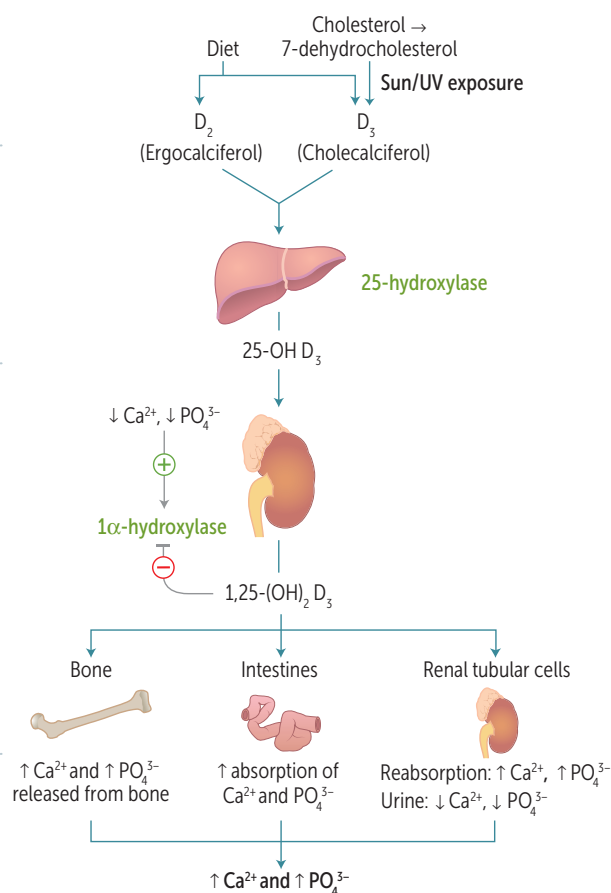
↑ PTH, ↓ Ca<sup>2+</sup>, ↓ PO<sub>4</sub><sup>3-</sup>  
 → ↑ 1,25-(OH)<sub>2</sub>D<sub>3</sub> production.  
 1,25-(OH)<sub>2</sub>D<sub>3</sub> feedback inhibits its own production.  
 ↑ PTH → ↑ Ca<sup>2+</sup> reabsorption and ↓ PO<sub>4</sub><sup>3-</sup> reabsorption in the kidney.

**DEFICIENCY**

Rickets in children (deformity, such as genu varum “bowlegs” **A**), osteomalacia in adults (bone pain and muscle weakness), hypocalcemic tetany.  
 Caused by malabsorption, ↓ sun exposure, poor diet, chronic kidney disease (CKD), advanced liver disease.  
 Give oral vitamin D to breastfed infants. Darker skin and prematurity predispose to deficiency.

**EXCESS**

Hypercalcemia, hypercalciuria, loss of appetite, stupor. Seen in granulomatous diseases (↑ activation of vitamin D by epithelioid macrophages).



Rx

**Vitamin E**

Includes tocopherol, tocotrienol.

**FUNCTION**

Antioxidant (protects RBCs and neuronal membranes from free radical damage).

**DEFICIENCY**

Hemolytic anemia, acanthocytosis, muscle weakness, demyelination of posterior columns (↓ proprioception and vibration sensation) and spinocerebellar tract (ataxia). Closely mimics Friedreich ataxia.

Neurologic presentation may appear similar to vitamin B<sub>12</sub> deficiency, but without megaloblastic anemia, hypersegmented neutrophils, or ↑ serum methylmalonic acid levels.

**EXCESS**

Risk of enterocolitis in infants (infants) with excess of vitamin **E**.

High-dose supplementation may alter metabolism of vitamin K–dependent proteins (factors II, VII, IX, X; protein C/S) → enhanced anticoagulant effects of warfarin.

**Vitamin K**

Includes phytonadione, phylloquinone, phytonadione, menaquinone.

**FUNCTION**

Activated by epoxide reductase to the reduced form, which is a cofactor for the  $\gamma$ -carboxylation of glutamic acid residues on various proteins required for blood clotting. Synthesized by intestinal microbiota; dietary sources include leafy greens.

**K** is for **K**oagulation. Necessary for the maturation of clotting factors II, VII, IX, X, and proteins C and S. Warfarin inhibits vitamin K–dependent synthesis of these factors and proteins.

**DEFICIENCY**

Neonatal hemorrhage with  $\uparrow$  PT and  $\uparrow$  aPTT but normal bleeding time (neonates have sterile intestines and are unable to synthesize vitamin K). Can also occur after prolonged use of broad-spectrum antibiotics or hepatocellular disease.

Not in breast milk; “breast-fed infants **D**on’t **K**now about vitamins **D** and **K**”. Neonates are given vitamin K injection at birth to prevent hemorrhagic disease of the newborn.

**Zinc****FUNCTION**

Mineral essential for the activity of 100+ enzymes. Important in the formation of zinc fingers (transcription factor motif).

**DEFICIENCY**

Delayed wound healing, suppressed immunity, male hypogonadism,  $\downarrow$  adult hair (axillary, facial, pubic), dysgeusia, anosmia. Associated with **acrodermatitis enteropathica** **A**—congenital defect in intestinal zinc absorption manifesting with triad of hair loss, diarrhea, and inflammatory skin rash around body openings (periorificial) and tips of fingers/toes (acral). May predispose to alcoholic cirrhosis.

**Protein-energy malnutrition****Kwashiorkor**

Protein malnutrition resulting in skin lesions, edema due to  $\downarrow$  plasma oncotic pressure (arising from  $\downarrow$  serum albumin and  $\downarrow$  antidiuretic hormone), liver malfunction (fatty change due to  $\downarrow$  apolipoprotein synthesis and deposition). Clinical picture is small child with swollen abdomen **A**.

Kwashiorkor results from protein-deficient **MEALS**:

**M**alnutrition

**E**dema

**A**nemia

**L**iver (fatty)

**S**kin lesions (eg, hyperkeratosis, dyspigmentation)

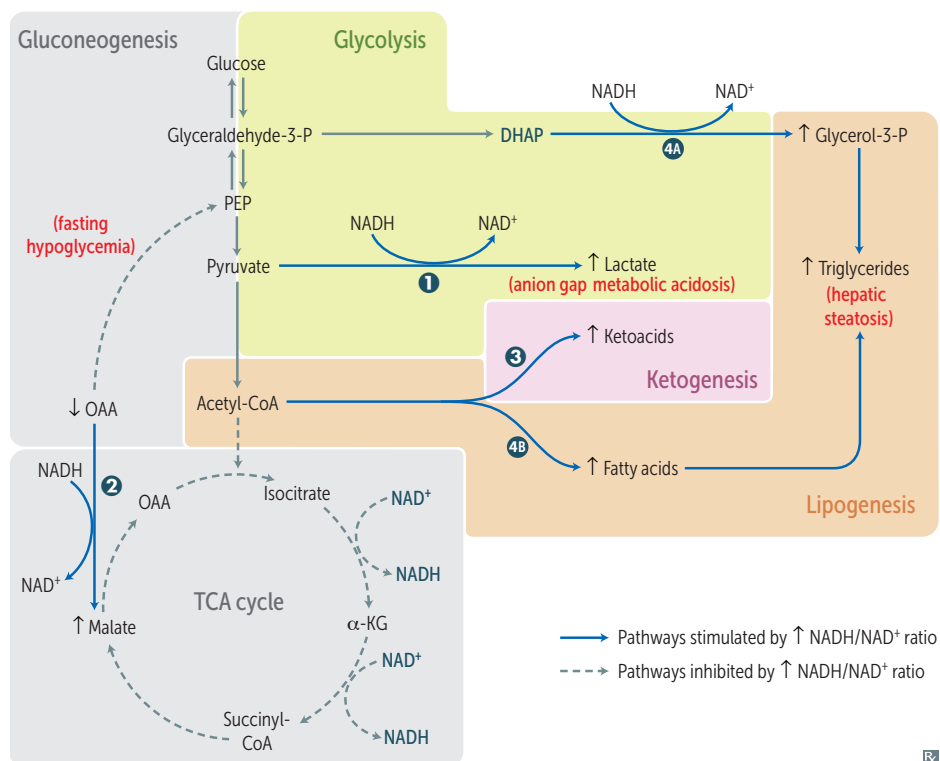
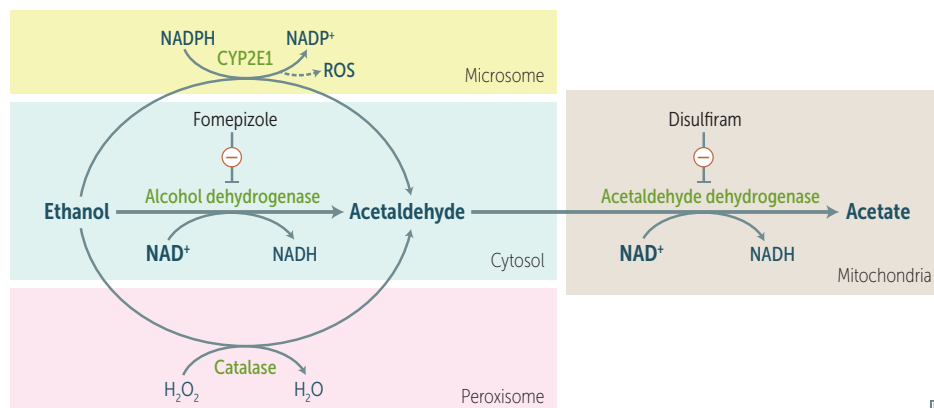
**Marasmus**

Malnutrition not causing edema. Diet is deficient in calories but no nutrients are entirely absent.

**M**arasmus results in **m**uscle wasting **B**. Linear growth maintained in acute protein-energy malnutrition (vs chronic malnutrition).



## Ethanol metabolism



↑ **NADH/NAD<sup>+</sup>** ratio inhibits TCA cycle → ↑ acetyl-CoA used in ketogenesis (→ ketoacidosis), lipogenesis (→ hepatosteatorosis). Females are more susceptible than males to effects of alcohol due to ↓ activity of gastric alcohol dehydrogenase, ↓ body size, ↓ percentage of water in body weight.

**NAD<sup>+</sup>** is the limiting reagent. Alcohol dehydrogenase operates via zero-order kinetics.

Ethanol metabolism ↑ **NADH/NAD<sup>+</sup>** ratio in liver, causing:

- 1 Lactic acidosis—↑ pyruvate conversion to lactate
- 2 Fasting hypoglycemia—↓ gluconeogenesis due to ↑ conversion of OAA to malate
- 3 Ketoacidosis—diversion of acetyl-CoA into ketogenesis rather than TCA cycle
- 4 Hepatosteatorosis—↑ conversion of DHAP to glycerol-3-P (4A); acetyl-CoA diverges into fatty acid synthesis (4B), which combines with glycerol-3-P to synthesize triglycerides

**Fomepizole**—competitive inhibitor of alcohol dehydrogenase; preferred antidote for overdoses of methanol or ethylene glycol. Alcohol dehydrogenase has higher affinity for ethanol than for methanol or ethylene glycol → ethanol can be used as competitive inhibitor of alcohol dehydrogenase to treat methanol or ethylene glycol poisoning.

**Disulfiram**—blocks acetaldehyde dehydrogenase → ↑ acetaldehyde → ↑ hangover symptoms → discouraging drinking.